Full Stack Assignment – 29

Assignment ID: FSD-29
Title: Subscription Control Management System
Duration: 3 Hours

1 Objective

Design and develop a **Subscription Control Management System** that allows users to subscribe to a service and administrators to manage subscriptions.
The system must enforce a single active subscription per user and ensure **data persistence using a database**.

2 User Roles

USER

- Register and log in to the system
- View available subscription plans
- Subscribe to a plan
- View subscription status

ADMIN

- Register and log in to the system
- Create and manage subscription plans
- View all user subscriptions

3 Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Subscription Management

- ADMIN can create subscription plans
- Subscription plan data must be stored in the database
- USER can subscribe to an available plan
- Subscription details must persist in the database

4 Business Rule (Critical)

⚠️ **A user can have only one active subscription at any given time.**

If a user already has an active subscription, new subscription attempts must be rejected.

5️⃣ Technical Constraints (Mandatory)

- All subscription and plan data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for USER and ADMIN
- Subscription plan listing and subscribe action for USER
- Subscription plan management and overview for ADMIN

UI design is not evaluated; rule enforcement and correctness matter.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions
- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8 Evaluation Note

Evaluation focuses on **constraint enforcement, role control, and end-to-end execution**.